第八届ACM-ICPC暨CCCC-GPLT校内选拔赛决赛题目原文

Problem A: Arrangement For Snacks-crazyX

Arrangement For Snacks

发布时间: 2017年12月6日 19:25 最后更新: 2017年12月9日 14:00 时间限制: 1000ms 内存限制: 128M

描述

As a sponsor of our USTB programming contest, CrazyX has many circumstances should be considered. Recently, he has found that the degree of snacks can be a serious factor (the degree refers to whether the snack is tasty or not). Because he is busy preparing contest, he wants you to help.

For some reasons, CrazyX does not like eating some snacks with great difference of degree continuously in a day.

In details, suppose the degree of a snack can be measured by a positive integer i. A bigger number means that the snack is more delicious. In CrazyX's requirement, he wants to eat K snacks with consecutive degrees every day. Formally speaking, if one day he ate K snacks, their degrees must be $x, x+1, \dots x+k-2, x+k-1$ respectively. It is because if there are two snacks with the same degree, that is not good due to Marginal Diminishing Effect, and if two snacks have a big gap in degree, you cannot make CrazyX satisfied.

CrazyX has a large number of snacks, and he knows all the degrees. Now, he wants to arrange these snacks so that he can be satisfied as long as possible. Please help him to count the maximum number of days which he can be satisfied.

输入

The first line of the input contains an integer $T(1 \le T \le 10)$, which is the number of test cases. Then T test cases follow and each test case consists of two lines.

The first line contains two integers N and K $(1 \le K \le N \le 1000)$, where N means the kinds of degrees, and K is the number of snacks that she want to eat in one day. The second line contains N integers, the i-th one of which is an integer $a_i(0 \le a_i \le 10^5)$ indicating the number of snacks whose degree are i.

输出

For each test case, the output should be a single integer in a line indicating the maximum number of days which can make crazyX satisfied.

样例输入1

```
3
3 2
1 2 1
6 3
1 5 3 4 7 6
9 4
2 7 1 3 6 5 8 9 4
```

样例输出1

```
2
5
6
```

提示

Marginal Diminishing Effect: 边际递减效应

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

比赛已经结束

Problem B: CrazyX and RSA-marryjianjian

CrazyX and RSA

发布时间: 2017年12月6日 21:19 最后更新: 2017年12月9日 11:43 时间限制: 1000ms 内存限制: 128M

描述

CrazyX is learning RSA. These days he is confused about how to calculate the modular multiplicative inverse of the public key exponent. He asks marryjianjian for help. He already knew how to decrypt:

$$m \equiv c^d \ (mod\ n)$$

m is what we need. d is what CrazyX is puzzled. c is encrypted. Now marryjianjian helped CrazyX to calculate the d. Could you help CrazyX to get m ?

输入

There are multilpe cases. For each case: there are three integers in one line, $c\,(1 \le c \le 100)$, $d(1 \le d \le 100)$ and $n(1 \le n \le 100000)$.

输出

Output m descripted above one line.

样例输入1

```
2 3 7
23 2 97
```

样例输出1

```
1
44
```

提示

$(a * b)\ \%\ n = (a\ \%\ n) * (b\ \%\ n)\ \%n$

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

提交代码

比赛已经结束

Problem C: Kindergarten-yuanzhaolin

Kindergarden

发布时间: 2017年12月6日 21:24 最后更新: 2017年12月9日 11:43 时间限制: 3000ms 内存限制: 128M

描述

In order to escape from the bullying and abuse of kindergarten teachers, the child crazyX decides to escape from the kindergarten!

Suppose the kindergarten is a grid of n rows and m columns, and we give it a pair of numbers for each cell $(i,j)(1 \le i \le n, 1 \le j \le m)$, which means that the position of the cell is in the ith row and the jth column.

At the time of $T = 0$, the initial position of crazyX is at the top left corner $(1,1)$, and he needs to move to the safe exit which locates at the bottom right corner of the position (n,m) in the kindergarten as soon as possible.

In order to monitor children's behaviors, in each row, there is a camera installed on the left side of the leftmost grid to monitor whether there is a child in the row. crazyX attends to escape from the kindergarten unconsciously. So, he hopes not to be captured by any camera during his escaping.

Because the nurses in the kindergarten damaged the monitoring disk accidentally, **the working state of each camera changes periodically**, namely that the camera on the i row has a working period (t_i). When it keeps shooting and recording for exact $1s$ and then,it will stop working for $(t_i - 1)$ s . So, in the time of $(t_i - 1)$ s, crazyX can appear in the i row. **No all of the cameras start its $1s$ shooting at the same time. For each camera, we suppose that the first shooting moment is $s_i(s_i < t_i)$ during its the first period after crazyX's moving.**

The more detailed information is as follows:

1. crazyX can move a unit of distance per second, namely that he can move from (x,y) to $(x+1,y)$, $(x-1,y)$, $(x,y+1)$, $(x,y-1)$, also can stay still.
2. CrazyX cannot move beyond the boundary of the kindergarten, that is, the location of crazyX (x,y) have to satisfy, $1 \le x \le n, 1 \le y \le m$ at any time.
3. It's safe for crazyX when he stands at (1,1) when T = 0.
4. The movement of crazyX and the switch of the working state of cameras are both instantaneous processes, and at this moment, it is allowed that crazyX is captured. For example, if the camera in line 3 were running during the time $t \in (4s, 5s)$, then crazyX would be allowed to leave the 3rd line at the 4th second or to line 3rd at the 5th second.
5. crazyX needs to reach (n,m) safely, that means, he couldn't be captured by the camera in nth line

at (n, m) either, though he has already touched the exit door.

Now crazyX wants you to help him calculate the minimum number of seconds that he can safely escape from the kindergarten. If there is no escaping route from the kindergarten, please output -1.

输入

The first line of the input contains an integer $T (T \leq 10)$, denoting the number of test cases.
In each case, the first line contains two integers $n (n \leq 200)$, $m (m \leq 200)$
In second line, n integers $t_1, t_2 .... t_n (2 \leq t_i \leq 6)$ are listed denoting the working period of each camera.
In third line, n integers $s_1, s_2 .... s_n (0 \leq s_i < t_i)$ are listed denoting the start time of each camera.

输出

For each test case, if crazyX can can use to escape from kindergraden, you should output the minimum time he needs, else you should just output -1.

样例输入1

```
2
2 5
3 3
0 1
2 5
2 2
1 0
```

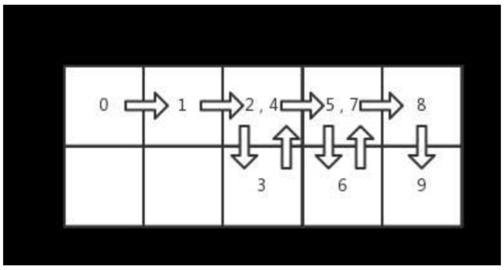
样例输出1

```
9
-1
```

提示

In the first case, crazyX can't stand in the 1st row when $T = 3, 6, 9$ and can't stand in the 2nd row at $T = 1, 4, 7$. One of the best routes is $(1,1) \rightarrow (1,2) \rightarrow (1,3) \rightarrow (2,3) \rightarrow (1,3) \rightarrow (1,4) \rightarrow (2,4) \rightarrow (1,4) \rightarrow (1,5) \rightarrow (2,5)$

.So the minimum time crazyX need is 9 seconds.



But in the second case, the working period of each cameras is too short to make the crazyX's escaping plan possible.

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

提交代码

比赛已经结束

Problem D: Laptops-Eopxt

Laptops

发布时间: 2017年12月6日 21:33 最后更新: 2017年12月9日 11:43 时间限制: 1000ms 内存限制: 128M

描述

One day nbyby and Smile had an argument about the price and quality of laptops. nbyby thinks that the more expensive a laptop is, the better it is. Smile disagrees. Smile thinks that there are two laptops, such that the price of one laptop is less (strictly smaller) than the price of the other laptop but the quality of the first laptop is higher (strictly greater) than the quality of the second laptop.

Please, check the guess of Smile. You are given descriptions of two laptops. Determine whose point of view is right.

输入

The input consists of T cases.
The first line contains a single integer T indicating the number of the cases. $(T \le 100)$
Then T lines, for each line there is a test case, contains 4 integers, a_1, b_1, a_2, b_2, indicating the price of the first laptop, the quality of the first laptop, the price of the second laptop, the quality of the second laptop. $(0 \le a_1, b_1, a_2, b_2 \le 100)$

Note that $a_1 \ne a_2$, $b_1 \ne b_2$.

输出

If Smile is correct, print "Happy Smile", otherwise print "Poor Smile" (without the quotes).

样例输入1

```
1
1 2 2 1
```

样例输出1

```
Happy Smile
```

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

比赛已经结束

Problem E: Oscar Buy an Apartment-marryjianjian

Oscar Buy an Apartment

发布时间: 2017年12月6日 21:36 最后更新: 2017年12月9日 11:43 时间限制: 1000ms 内存限制: 128M

描述

Oscar wants to buy an apartment in a new house at Line Avenue of LETTers. The house has n apartments indexed from 1 to n. The n apartments are in a row and two apartments are adjacent if their indexes number differ by 1. Some of the apartments may already be sold.

Oscar likes visiting his neighbors, so he wants to buy the apartment which is adjacent to at least one already-sold apartment . Oscar knows that the number of apartments already sold is k, but he doesn't know their indexes yet. Please find out the the minimum possible and the maximum possible number of apartments that Oscar is satisfied with.

输入

There are multiple cases. For each case: there are two integers, $n\,(1 \le n \le 1e9)$ and $k\,(0 \le k \le n)$, in one line.

输出

For each case print the minimum possible and the maximum possible number of apartments that Oscar are satisfied with in one line.Two numbers separated with a space.

样例输入1

```
6 3
```

样例输出1

```
1 3
```

提示

In the sample case above, if apartments numbered 1, 2 and 3 are bought by others, he only can buy the apartment 4. If apartments numbered 1, 3 and 5 are bought by others, he can choose an apartment among the apartments numbered 2, 4 and 6.

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

比赛已经结束

Problem F: Random Statistic-yuanzhaolin

Random statistics

发布时间: 2017年12月6日 21:40 最后更新: 2017年12月9日 11:44 时间限制: 1000ms 内存限制: 128M

描述

The Sports Department of USTB wants to statistic the height of students, and they currently collect a sample set of the height of n students. Because people usually make fun of the old saying "my math is taught by PE teachers", the PE teachers want to take this opportunity to show their math abilities and they put forward a problem as follows:

At first, let's define the values of three different properties of set S as A, B, C:

Assuming that the elements in S are $x_1, x_2 \ldots . x_{|S|}$, then the values of A, B, C are calculated as follows:

$$A = \sum_{i=1}^{|S|} x_i$$

$$B = (\sum_{i=1}^{|S|} x_i)^2$$

$$C = \max_{i=1}^{|S|} x_i$$

Specifically,$A = B = C = 0$ when the set S is empty.

Teachers want to know that if they randomly choose some values from the original data set(Each element was chosen by 50% Probality) to form a new set S (S can be empty), then what is the expectation of the values A, B, C of the collection?

PE teachers think the problem is too simple for themselves, so they want to take an examination of the smart you!

输入

The first line of the input gives the number of test cases T.
Each case begins with one line including an integer $n (1 \leq n \leq 1000)$: the size of the data set.
The next line contains n integers, the i-th of them is the value of the i element $x_i (0 \leq x_i \leq 1e^4)$in the data set.

输出

In each case, please print the values of A, B, C which are separated by space with exactly four digits after the decimal point (the number should be rounded to the fourth digit after the decimal point) and arranged in a single line.

样例输入1

```
2
2
1 2
3
3 3 3
```

样例输出1

```
1.5000 3.5000 1.2500
4.5000 27.0000 2.6250
```

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

提交代码

比赛已经结束

Problem G: Geometry-crazyX

Geometry

发布时间：2017年12月6日 21:05 最后更新：2017年12月9日 11:44 时间限制：1000ms 内存限制：128M

描述

The description is very simple.

Can you find $\mathbf{n+1}$ **different** points in a n-dimension space that the vectors from the origin point - $(0,0,0,\cdots,0)$ to each point are of the same length and the angle of every pair of vectors are equal.

Tips:

We can use $(x_1,x_2,\cdots,x_n)$ to describe a point and $(y_1-x_1,y_2-x_2,\cdots,y_n-x_n)$ to describe a vector from point $(x_1,\cdots,x_n)$ to point $(y_1,\cdots,y_n)$ in a n-dimension space.

The length of a vector $\vec{p}=(x_1,x_2,\cdots,x_n)$ is $|\vec{p}|=\sqrt{x_1^2+x_2^2\cdots+x_n^2}$.

The angle of vector $\vec{p}=(x_1,x_2,\cdots,x_n)$ and $\vec{q}=(y_1,y_2,\cdots,y_n)$ is $\theta=arccos(\frac{\vec{p}\cdot\vec{q}}{|\vec{p}|\cdot|\vec{q}|})$, of course $\vec{p}\cdot\vec{q}=x_1\cdot y_1+x_2\cdot y_2+\cdots+x_n\cdot y_n$.

输入

The first line contains a single integer $T\,(T \le 1000)$ means the test cases.

For each test case, the are a single integer $n\,(n \le 1000)$ means you need to find $\mathbf{n+1}$ points that satisfy the condition described above.

输出

For each test case you only need to output the $\frac{1}{cos\theta}$, θ means the angle of any two vectors. OBVIOUSLY the answer will be an integer.

样例输入1

```
1
2
```

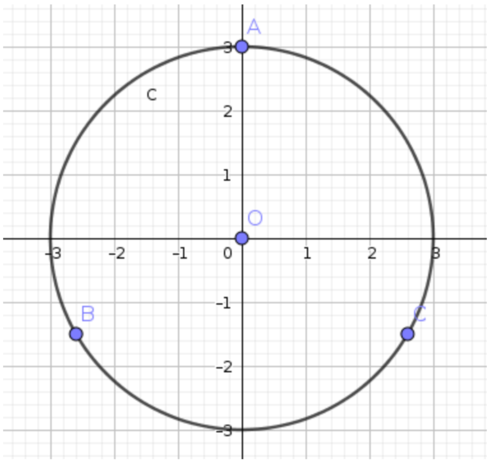
样例输出1

```
-2
```

提示

The sample explanation:

when $n = 2$ you can consider in the 2D plane and draw a cicle:



Obviously $|OA| = |OB| = |OC|$ and $\angle AOB = \angle AOC = \angle BOC = 120^{\circ}$

So the answer for $n = 2$ is $1/cos(120^{\circ}) = -2$

选择语言

◉ C (GCC 5.4) ○ C++ (G++ 5.4) ○ Java (Oracle JDK 1.8)

提交代码

提交代码

比赛已经结束

Problem H: Stones-owly

Stones

发布时间：2017年12月6日 21:42　最后更新：2017年12月9日 11:44　时间限制：4000ms　内存限制：128M

描述

There is a grid with $n \times m$ cells . CrazyX likes to place stones into it. There are some cells are empty and some are placed with at least one stone as he placed them randomly. But CrazyX wants to make sure there are at least one stone in each cell so he will move the stones. He wants to know whether he can achieve the situation described above by unlimited number of operation of two types only:

- choose a stone and move it right if it will not get out of the grid.

- choose a stone and move it down if it will not get out of the grid.

输入

There is a single integer T represents the number of test cases.

For each test case, there are two integers n and m $(1 \le n, m \le 800)$ means the size of the grid and a integer $k (k \le 10^6)$ means the number of stones in the first line. Following are k lines. The ith line contains two integers x_i, y_i $(1 \le x_i, y_i \le 800))$ means the position of the ith stone.

输出

Output YES if CrazyX can do it or NO if he can not.

样例输入1

```
2
2 2 4
1 1
1 1
1 1
1 1
2 2 4
1 1
2 1
2 1
2 2
```

样例输出1

```
YES
NO
```

样例输入2

```
1
2 2 4
1 1
1 1
2 2
2 2
```

样例输出2

```
NO
```

提示

PLEASE use scanf to read the data for the input file is too large.

选择语言

C (GCC 5.4)　C++ (G++ 5.4)　Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

比赛已经结束

Problem I: Summer Social Practice of Electronic Department of SUOAO-Martian

Practice of SUOAO

发布时间：2017年12月6日 21:50 最后更新：2017年12月9日 11:44 时间限制：1000ms 内存限制：128M

描述

Summer vacation is coming! But freshmen are going to do their social practice.

SUOAO is a technology club under School of Automation and Electrical Engineering(SAEE). Every year it will organize two social practice project. One is for Electronic Department and another is for Software Department. Electronic Department will make an electronic product to achieve a specific function which have some social meanings. You can finish your social practice meanwhile learning a lot of electronic knowledge, which is a very rich experience.

Now the members of Electronic Department are designing the electronic product of their social practice. Considering social meanings, the product should not be expensive ,bulky or heavy.

To make the problem simple, we defined every type of component a bulk, cost and value. Here are some type of component, the quantity we use one type of component has no limit, but we should considering the bulk limit and the cost limit. Giving some component property and limitation, calculate the most optimal purchase scheme with the maximum value.

输入

There is a integer $T(1 \le T \le 10)$ means the number of test cases in the first line.

For each test case, the first line contains three integers $N(1 \le N \le 100), B(1 \le B \le 30), C(1 \le C \le 300)$ means the number of component types, the bulk limitation and the spend limitation. Each of the next N lines contains name, bulk($b_i(1 \le b_i \le 10)$), cost($c_i(1 \le c_i \le 100)$), value($v_i(1 \le v_i \le 10000)$). The length of name is between 2 and 11.

输出

For each test case, output the maximum value we can get from the component list.

样例输入1

```
1
5 30 300
atmega16 4 140 750
74hc595 2 45 200
led 2 1 60
buzzer 7 12 380
adconverter 3 46 265
```

样例输出1

```
2355
```

提示

In sanple case, the component we chosen is:

74hc595 1

adconverter 1

atmega16 1

buzzer 3

The value is 1*200+1*265+1*750+3*380=2355.

选择语言

C (GCC 5.4) C++ (G++ 5.4) Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

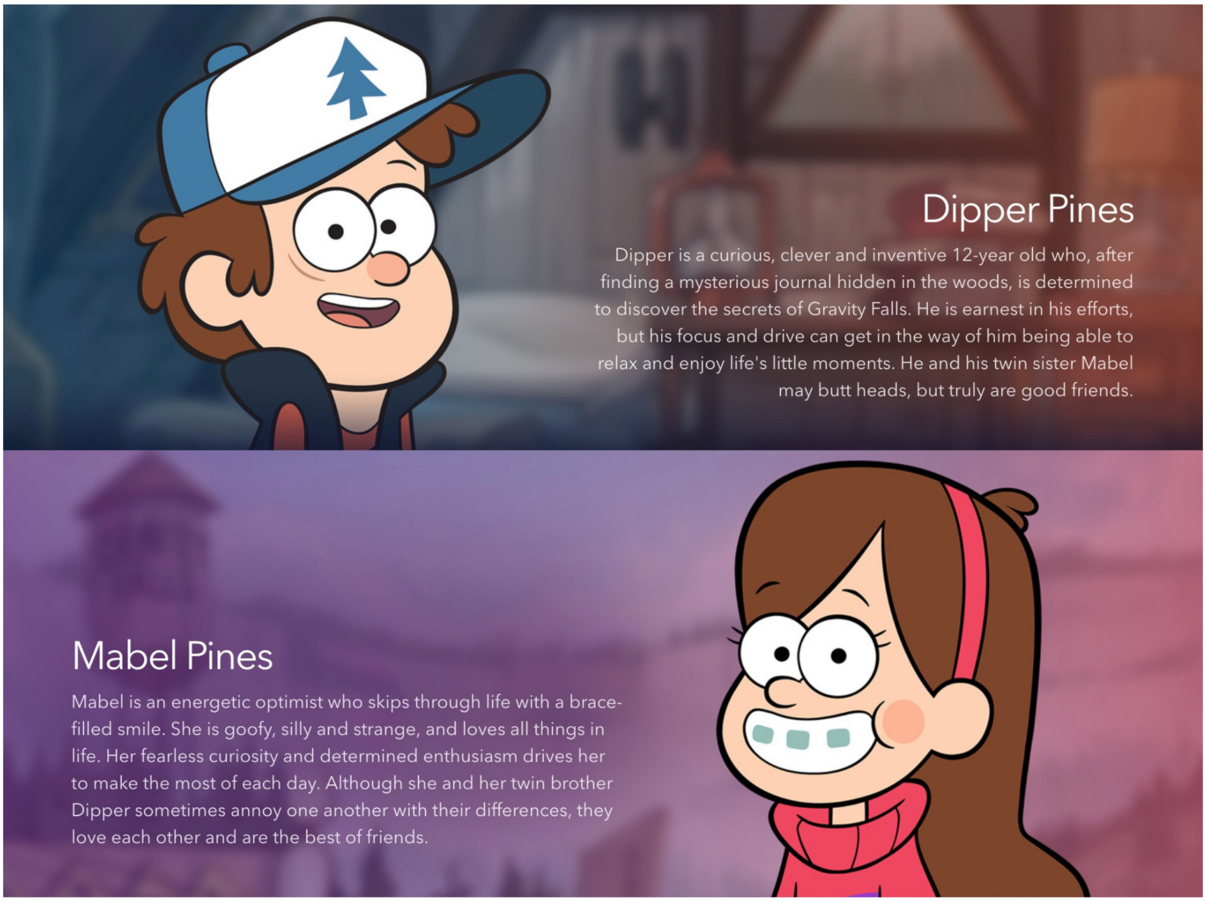
比赛已经结束

Problem J: Dipper and Mabel-smile & nbyby

Dipper and Mabel

发布时间：2017年12月7日 13:44 最后更新：2017年12月9日 11:44 时间限制：1000ms 内存限制：128M

描述



Have you ever heard of the old legendary story of Dipper and Mabel? The story goes that for their summer vacation, 12-year-old Dipper Pines and his twin sister Mabel Pines are dropped off from their home in Piedmont to the miraculous town of Gravity Falls to spend the summer with their uncle Grunkle Stan, who runs a tourist trap called the 'Mystery Shack'. Things are not what they seem in this small town, and with the help of a mysterious journal that Dipper finds in the forest, they begin unraveling the local mysteries. However, it may disappoint you with no journal and little mysteries, because there will only be a party today at Gravity Falls. Oh, please, don't put on such an unhappy face! The party is also special and won't let curious guys down! It's a twin's party!

A twin's party means that all the $N*2$ participants are all twins, N pairs of twins, just like Dipper and Mabel. As we all know, twins are always alike and it's hard to distinguish them. So, there will be a name-card-exchanging game for the twins to get acquainted with each other. When one of a pair of twins, for example Dipper, meet someone, for example Stanley from some other pair of twins, they'll exchange their name cards. That is, Dipper will get a name cards of Stanley and Stanley will also get one of Dipper. Obviously, your twin sister/brother is usually the most familiar person on the earth for you, so twins will never exchange name cards with his brother/her sister. And two persons will exchange at most once.

After a while, Mabel got tired of the name-card-exchanging party and went for a date with her new boyfriend. Dipper also got tired of the game, but he found something really mysterious! The $N*2-1$ participants left at the party have different number of new-friend name cards! That's to say, everyone except Mabel has received A_i name cards, and these $N*2-1$ A_is are all different! It can't be just a coincidence! Will Mabel be a part of this miracle? Now Dipper want you to tell how many new-friend name cards Mabel has received.

输入

The input consists of several test cases.
The first line contains a single integer T indicating the number of test cases. $(T \leq 15)$
Then T lines, for each line there is a test case, contain an integer, N $(0 \leq N \leq 10)$.

输出

For each test case, print the number of new-friend name cards Mabel received in one line.

样例输入1

```
1
2
```

样例输出1

```
1
```

选择语言

◉ C (GCC 5.4) ○ C++ (G++ 5.4) ○ Java (Oracle JDK 1.8)

提交代码

```
1
```

提交代码

比赛已经结束